```
#include<iostream>
using namespace std;

class node{
   public:
   int data;
   node*next;
   node(int x)
   {
      data=x;
      next=NULL;
   }
};
class queue{
   private:
   node*front;
   node*rear;

   public:
   queue(){
      front =rear=NULL;
   }
   int empty(void)
   {
      if(front == NULL)
      {
         return 1;
      }
      else{
         return 0;
      }
   }
   int peek(void)
   {
      if(front == NULL)
      {
         return 1;
      }
      else{
         return 0;
      }
   }
   void enqueue(int x)
   {
      node*p=new node(x);
      if(front == NULL)
      {
         front =p;
```

```
            rear=p;
        }
        else{
            rear->next =p;
            rear=p;
        }
    }

     int dequeue(void)
    {
        if (empty())
        {
            cout << "\nQueue is empty";
            return -1;
        }

        int x = front->data;
        node *p = front;

        front = front->next;

        if (front == NULL)
        {
            rear = NULL;
        }

        delete p;

        return x;
    }

    void display(void)
    {
        node *q = front;

        cout << "\nQueue elements: ";

        while (q != NULL)
        {
            cout << "\t" << q->data;
            q = q->next;
        }
    }
};

int main()
{
    int x;
```

```
    queue q;

    q.enqueue(33);
    q.enqueue(84);
    q.enqueue(112);
    q.enqueue(57);
    q.enqueue(90);
    q.enqueue(73);

    q.display();

    if (!q.empty())
    {
        x = q.dequeue();
        cout << "\nRemoved: " << x;
    }

    q.display();

    x = q.peek();
    cout << "\nFront element is: " << x;

    return 0;
}
//[Running] cd "c:\Users\STUDENT\Desktop\manasi kulkarni 209\" && g++ queueusingll.cpp
-o queueusingll && "c:\Users\STUDENT\Desktop\manasi kulkarni 209\"queueusingll

//Queue elements:     33      84      112     57      90      73
//Removed: 33
//Queue elements:     84      112     57      90      73
//Front element is: 0
//[Done] exited with code=0 in 0.356 seconds
```